Technology Stack

Date	12 July 2026
Team Id	XXXXX
Project Name	Opti Crop: Smart Agricultural Production Optimization Engine
Maximum marks	2 marks

Define your technology Stack:

Identify the programming languages, frame works, databases, frontend tools, back end tools and API s that you will use to build your solution .A well-chosen technology stack ensures that your application Is scalable, maintainable and aligned with your project requirements

SNO	Architecture /Component	Technology chosen	purpose
1.	Frontend / Client-Side	HTML, CSS, JavaScript, Bootstrap	Creates a responsive and user-friendly interface for farmers to enter data and view recommendations.
2.	Backend / Server-Side	Python, Flask	Handles user requests, processes data, and connects the ML model with the web application.
3.	Database / Data Storage	CSV Dataset, Pandas	stores crop, soil, and weather data used for prediction and analysis.
4.	Machine Learning	Scikit-learn (Logistic Regression), NumPy	Builds and trains the crop prediction model for accurate recommendations.

5.	Cloud / Hosting / Deployment	Localhost (Flask) / Render	Runs and deploys the application for easy access
6.	Version Control & CI/CD	Git, GitHub	Manages source code, tracks changes, and enables collaboration.
7.	Third-Party APIs / Other Tools	Open Weather API	Provides real-time weather data for accurate crop recommendation